Exp. 9

Implement subnetting in cisco packet tracer simulator

Aim :-

To implement subnetting in cisco packet tracer simulator by designing multiple subnets with suitable number of hosts, assigning static IP addresses, connecting subnets through router and simulating packet transmission.

Procedure :-

1. creating a network topology
   - open cisco packet tracer & select new from top left corner
   - select network → generic to create blank network topology
   - Add the required device to network topology
   - Devices used are routers, switches, PCs
   - To add device, select required device from Bottom left corner & drag it to network topology
   - connect devices by dragging cable from one device's port to another device port.

2. Subnetting

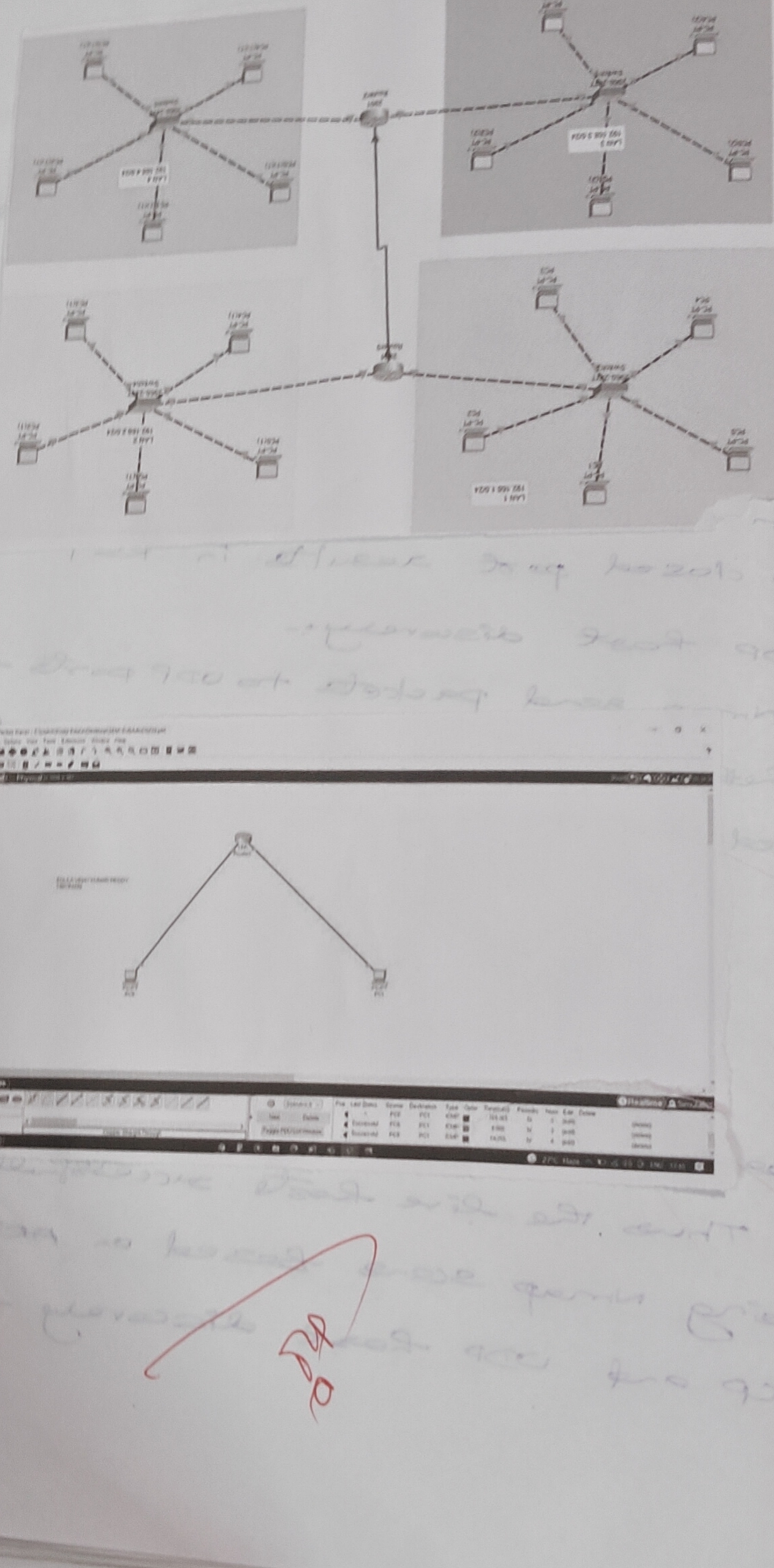
   The network address 192.168.0.0/24 is subnetted to provide enough space for at least 5 address for end devices, switch and router.

   A /27 subnet mask is used

   The /27 subnet mask provides :

   8 subnets

   30 host address per subnet



commands -

enable

configure terminal

interface FastEthernet 0/0

ip address & IP address y & subnet mask

no shutdown

exit

interface Fast ethernet 0/1

ip address & IP address y & subnet mask

no shut down

exit



3. configuring the devices
   After adding & connecting devices, configure the router.
   Right click on router & select CLI.

4. configuring the switch

5. configuring the PCs

6. configuring GigaBitEthernet interface

7. Testing the network
   open command prompt on each PC
   use the ping command
   observe the packet transmission

Result :-

using cisco packet tracer with multiple subnets created using a /27 subnet mask, static IP address assigned to devices & connectivity tested through router.